Physics 3200Y: Lab 2

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Abstract

Blah balh

1 Introduction

This lab seeks to develop techniques to calculate capacitance of a constructed capacitor at varying frequencies and configurations. Additionally, through relationships derived in this report, the properties of the dielectric used in the construction of the capacitor are determined. Several new-to-the-authors techniques were developed through the course of this lab such as Fourier transforms.

2 Theory

Maybe squeeze a derivation the expression for phase in here somewhere. See 2b?

2.1 Sawyer-Tower Circuit

The Sawyer-Tower circuit was used to facilitate data collection on the constructed capacitor.

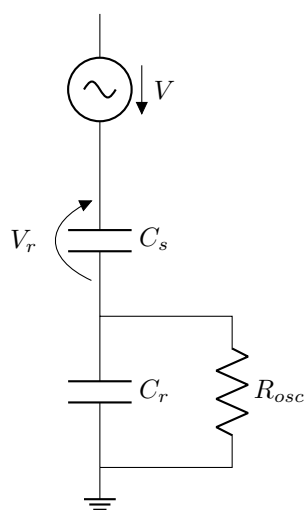

Figure 1: The Sawyer-Tower circuit. C_s is the sample capacitor whose properties are to be investigated, C_r is a reference capacitor of known parameters. R_{osc} is **something?? idk what**

Do the derivation for the capacitance

Maybe some explanation

2.2 Complex Capacitance

2.3 Fourier Transform

Note that we just used numpy's r(eal)fft implementation. We may still want to sketch out the tutorial we did on it though.

3 Methods

3.1 Capacitor Construction

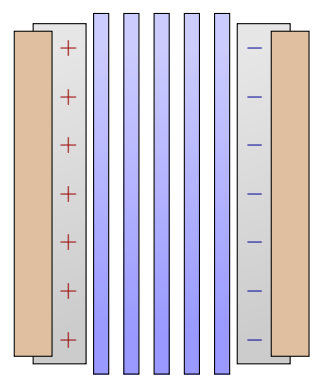

Figure 2: A cartoon of the constructed capacitor. The dielectric sheets (in blue) are represented with exaggerated spacing.

3.2 Data Collection

4 Results

The capacitance of the constructed capacitor was successfully determined using data collected from the Sawyer-Tower circuit and numpy's rfft. From this the dielectric "constant" was calculated using equation and found to vary not insignificantly with frequency. This is consistent with real-world data collected on the dielectric *functions* $\epsilon(\omega)$ of many materials, even those designed to have similar dielectric/optical properties over large frequency ranges.

cite!

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4.1 Capacitances & Resistances

Legends

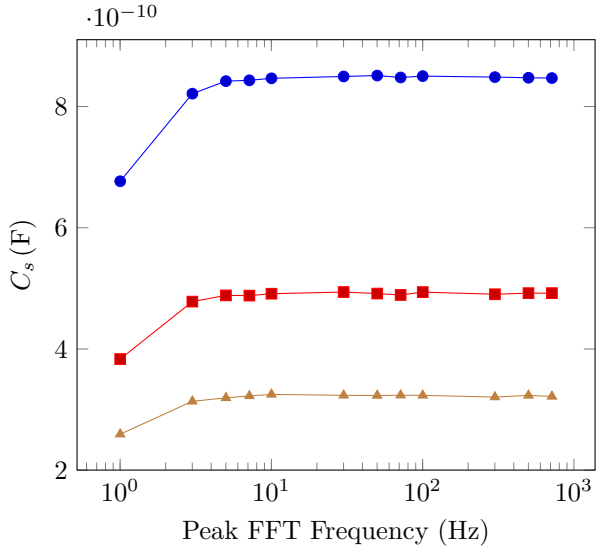

Figure 3: Recovered capacitances vs. recovered FFT peak frequencies

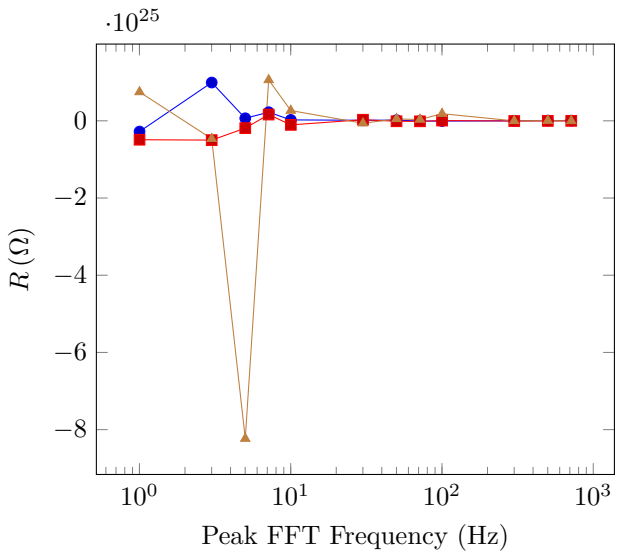

Figure 4: Recovered resistances vs. recovered FFT peak frequencies

4.2 Dielectric Function

The dielectric function was successfully calculated using the derived formula. As can be seen in 5 is sits between 2.4 and 2.8 (in units of ϵ_0) which is a reasonable range for common plastics which generally sit between 2 and 5 [1]

ref

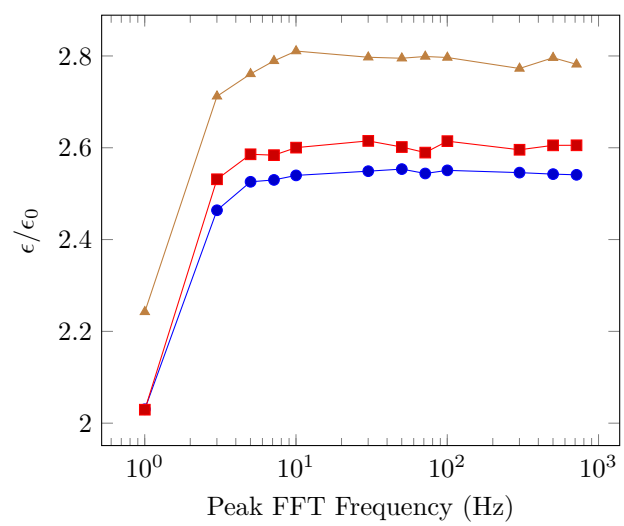

Figure 5: ϵ/ϵ_0 vs. recovered FFT peak frequencies

4.3 Fourier Transform

A Fourier Transform was successfully applied to the gathered data using Numpy's builtin `numpy.fft.rfft` function. As discussed in section 2.3 the Fourier Transforms are used to calculate the capacitance and thereby dielectric function. The actual effect of the transform is not very evident there, so it is discussed in more detail here.

Make sure it is discussed

v As seen in 6 the FFT is able to very effectively recover peak frequencies the collected V_r voltage data. There are however several remarkable features of this plot.

Recovered Frequency (Hz)	FWHM
1.0	1.03
3.0	1.03
5.01	1.02
7.15	1.27
10.01	1.02
30.04	1.03
50.07	1.02
71.53	2.18
100.14	1.02
300.43	1.03
500.72	1.02
715.31	2.2
1001.43	1.02

Table 1: FWHM of coefficient peaks at different recovered FFT frequencies.

Most immediately noticeable is the significant widening of the frequencies which are multiples of 7. As seen in 1 the FWHM for frequencies that are multiples of 7 is more than double the FWHM of other frequencies (the lower value at 7.15 Hz seems to be consequence of how Scipy's peak detection works. It is evident on the plot that this peak is also much wider than others). It is unclear why this is, literature available from a cursory search does not mention this, so it is likely something to do with the experimental configuration (Maybe some kind of resonant reflection as the widening is also present in the FFT of V 7) rather than the methods by which the data is processed. It is not of great consequence in this lab however as we are only interested in the coefficient values at the peak FFT frequencies, though it is an interesting phenomenon that might warrant further investigation.

The second interesting feature of 6 is the noticeable bumps at frequencies below 1 Hz in the frequencies less than 7 Hz. This again is not of consequence for this lab but is interesting as these low frequencies were not emitted by the function generator (as can be seen in **??**) and so are likely some kind of interference picked up by the constructed capacitor.

5 Discussion

6 Conclusion

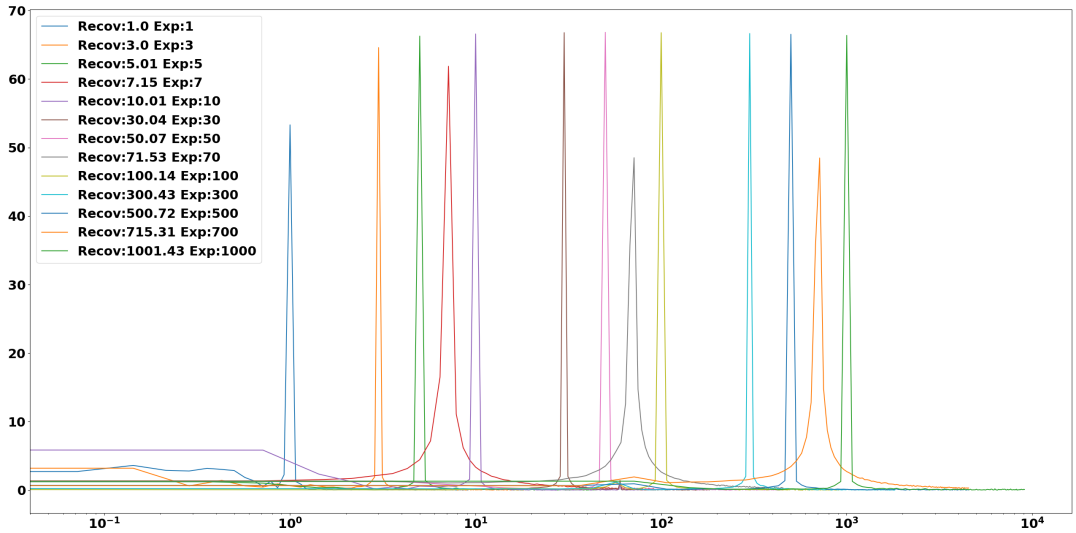

Figure 6: FFT coefficients vs. Frequency from V_r.

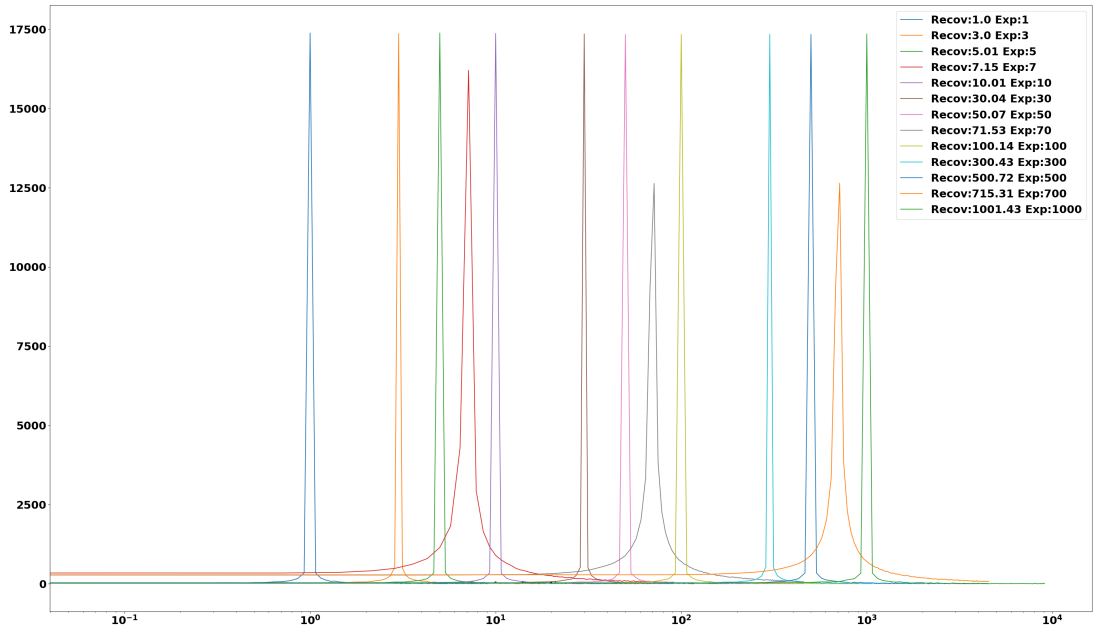

Figure 7: FFT coefficients vs. Frequency from V, the signal directly from the function generator.

7 References

[1] VEGA Grieshaber KG. Dielectric constants of plastics. Online, 3 2013.